Information Retrieval

Exercise – Winter term 2026/2027

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Agenda

1. Organization

2. Lab Projects

3. Next Steps

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Organization

Communication

- Slides, Announcements & Materials on the course website [temir.org]
- Questions and communication via Discord and email [discord.gg]
  - Category “Group Channels WiSe 2026”, channel “uk-ir-ws26”
  - Please use your full name as the server nickname



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Organization

Lab / Exercise

- Joint collaborative exercises with Friedrich Schiller University Jena
- Lab project as group work (3 people per group)
- Lab/ theoretical exercise sessions throughout the semester
  - **Tuesday, 14:00 – 15:30, Room 1114**
  - Lab
    - Each group shortly presents their current progress
    - Discuss open questions and next steps
  - Exercise
    - Exercises for lecture contents
    - Find (additional) notebooks/ tutorials on `teaching-ir-with-shared-tasks`
- We are also available for questions outside the scheduled times via Discord
  - Feel free to use the discussion board to discuss across universities!

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Lab Project

Overview

- Goal
  - Build & evaluate an information retrieval system for web search
  - Search for related work, learn data handling, indexing, select & implement suitable retrieval models, evaluate search effectiveness
  - Identify (and solve) retrieval errors
  - Submit written report and software on the TIRA platform

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Lab Project

Overview

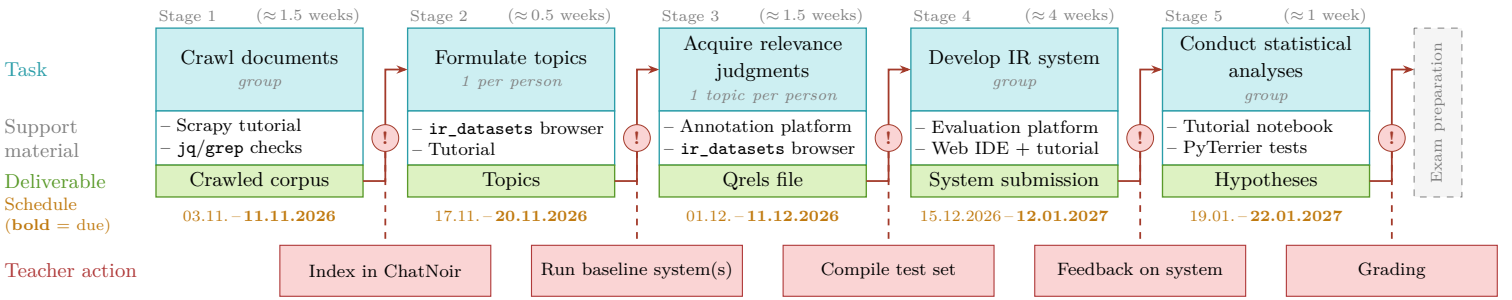
- Goal
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- Resources
  - TIRA as an evaluation platform & leaderboard [website, paper]
  - `ir_datasets` browser [website]
  - Experimentation with PyTerrier [paper]
  - Tutorials [IR, crawling]

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Lab Project

Overview



Overview of the course stages.

- Lab organized in 5 stages throughout the semester
- Each stage is finished by a deliverable
  $\rightarrow$ Groups hand in their intermediate work
- Deliverables are exchanged with other universities

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Lab Project

Task & Data

- Task: Web search
  - Aka: What you search everyday...
  - Very broad set of topics
- Crawled data
  - Scrapy tutorial
  - Collect data with upcoming tasks in mind
- Further evaluation data will be collected in the lab itself

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Lab Project

Stage 1: Crawl data

- Use Scrapy to crawl a data corpus for your search task
- Resulting jsonl file size: 400–500 MB of data per person
- Due Date: Wednesday, 11.11.2026, 23:59
- Deliverable: data corpus submitted online

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Crawl data

Scrapy

- Spider: [documentation]
  - Python class
  - defines where to start (`start_urls`)
  - defines how to parse a page (`parse`)
- Requests & responses:
  - Scrapy downloads pages asynchronously (many at once)
  - Each response page passes to `parse` callback
- Selectors: extract data from the HTML [documentation]
- Items: each yielded dict becomes one JSON line [documentation]
- Settings: control speed and politeness, e.g. `DOWNLOAD_DELAY`, `AUTOTHROTTLE_ENABLED`, `ROBOTSTXT_OBEY` [documentation]

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Crawl data

Scrapy Example

- framework for crawling websites and extracting structured data [website]
- cf. example from their documentation:

```
import scrapy

class QuotesSpider(scrapy.Spider):
  name = “quotes”
  start_urls = [“https://quotes.toscrape.com/tag/humor/”,]

  def parse(self, response):

      for quote in response.css(“div.quote”):

          yield {

              “author”: quote.xpath(“span/small/text()”).get(),
              “text”: quote.css(“span.text::text”).get(),
          }
      next_page = response.css(‘li.next a::attr(“href”)’).get()

      if next_page is not None:
          yield response.follow(next_page, self.parse)
```

(1) Spider name: unique identifier.
(2) Start URLs: first pages to crawl.
(3) Parse method: called for each downloaded page.
(4) Selectors (CSS or XPath): pick elements from the page.
(5) Yield: emits one item (a dict) → one line in the JSONL.
(6) Next page: extracts the pagination link.
(7) Follow link: requests the next page and parses it the same way.

- run the spider:

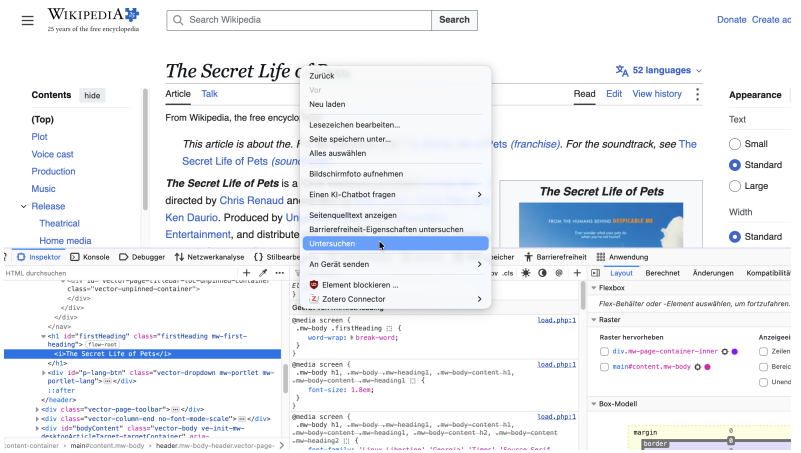
```
scrapy runspider quotes_spider.py -o quotes.jsonl
```


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Crawl data

Scrapy

1. Choose website(s) that fit your search task
2. Obey `robots.txt` and terms of use, e.g., Wikipedia's robots.txt
3. Find the elements you need with the browser's developer tools:



4. Write the spider: yield one document per page (`url`, `title`, `text`)
5. Test small first
   - settings: `CLOSESPIDER_ITEMCOUNT=50`
   - check output
   - scale to 400–500 MB

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Lab Project

Stage 2: Formulate topics

Create topics for the supplied search task and data collection.

- A topic is a description of a user's information need [lecture 2025: Topics]
  - A *text* that could be entered into the search system as a query
  - A precise *description* of the underlying information need
  - A *narrative* describing what is relevant to the topic (and what not)
- Due Date: Friday, 20.11.2026, 23:59
- Deliverable: valid topics submitted online

Topic ID	1
Text	retrieval system improving effectiveness
Description	What papers focus on improving the effectiveness of a retrieval system?
Narrative	Relevant papers include research on what makes a retrieval system effective and what improves the effectiveness of a retrieval system. Papers that focus on improving something else or improving the effectiveness of a system that is not a retrieval system are not relevant.

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Lab Project

Stage 3: Acquire relevance judgments

Judge the relevance of documents retrieved for your topic.

- Given your topics, we will retrieve documents
  - Top-10 for each topic and baseline system [lecture 2024: Pooling]
  - Some manually added documents
- Annotate these documents w.r.t. relevance to the topic (read the *narrative*)
- Due Date: Friday, 11.12.2026, 23:59
- Deliverable: Relevance judgments finished on Doccano

Topic ID	1
Query	retrieval system improving effectiveness
Topic desc.	What papers focus on improving the effectiveness of a retrieval system?
Topic narrative	Relevant papers include research on what makes a retrieval system [...]
Document ID	2005.ipm_journal-ir0anthology0volumeA41A1.7
Document text	In this paper we will present a language-independent probabilistic [...]
Relevance	1 (relevant)

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Lab Project

Stage 4: Develop information retrieval system

Build and evaluate your own IR system using
your topics and relevance assessments.

- Implement your IR system
  - Training data will be supplied & compute resources available
  - Final system deployed to the TIRA platform
- Evaluate your IR system
  - Topics from Open Web Index and TREC are used for validation
  - Tests carried out on the TIRA platform
- Summarize your system and findings in a preliminary report
- Due Date: Tuesday, 12.01.2027, 23:59
- Deliverable: Preliminary report (max. 2 pages), TIRA submission

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Lab Project

Stage 5: Conduct statistical analyses

Derive and test hypotheses with your previously constructed IR system.

- Look at types of retrieval errors
- Observe your system's behavior compared to baselines
- Formulate 2 hypotheses
- Hypotheses must be testable (more on that later)
- Due Date: Friday, 22.01.2027, 23:59
- Deliverable: Report (max. 2 pages)

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Questionnaire

Raise your hand if you...

- ... have used Python before?
- ... have previous knowledge in ML?
- ... have worked on data analysis before?
- ... have done data annotation before?
- ... have done scientific writing before?
- ... have done statistical analyses before?
- ... have used Docker before?
- ... would like to participate in an international shared task?

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Next Steps

Assignment

- Exercise sheet on the website [temir.org]
- Form groups
  - 2-3 people
  - Each group should have a unique group name (be creative!)
  - Deliverable: Email me the names of all group members, the name of your group, and Discord handles
  - Due Date: Friday, 06.11.2026, 23:59
- Crawl & submit your own crawled data [form]
  - One crawl per team member
  - Individual submission
  - Deliverable: crawled data submitted online
  - Due Date: Wednesday, 11.11.2026, 23:59

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